

VAPO: Value-Model-Based RL for Advanced Reasoning

Achieves 60.4 on AIME 2024, Surpassing
DAPO by 10+ Points

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Base Model: Qwen2.5-32B (No SFT Data)

Why Value Models? Finer Credit Assignment for Long CoT

- Limitations of Value-Model-Free Methods (GRPO, DAPO):
 - Compute advantage based on final trajectory reward and assign it uniformly across all tokens.
- Value Models Enable Precise Token-Level Credit Assignment:
 - Tracing Impact: Pinpoints each action's effect on returns, which is critical when subtle step-level errors cause catastrophic failures in multi-step reasoning.
 - Variance Reduction: Provide lower-variance estimates per token compared to Monte Carlo methods.
 - Generalization: A well-trained value model

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

- Training value models for long-CoT tasks faces three core hurdles:
- **1. Value Model Bias**
 - Arises from long trajectories and bootstrapped learning, leading to significant initial bias.
- **2. Heterogeneous Sequence Lengths**
 - Variable response lengths require different bias-variance tradeoffs (short responses need lower variance; long responses need lower bias).
- **3. Reward Signal Sparsity**
 - Only the final <eos> token receives a reward, an issue highly exacerbated by long chain-of-thought generation.

VAPO Framework: Integrated Solutions for Value-Model-Based RL

- VAPO integrates multiple techniques to map directly to core challenges:
- **Addressing Value Model Bias:**
 - Value-Pretraining: Pre-trains the value model before RL to significantly reduce initial bias.
 - Decoupled-GAE: Separates GAE computation to handle value estimation robustly.
- **Addressing Heterogeneous Lengths:**
 - Length-Adaptive GAE: Dynamically adjusts λ parameter based on response length.
- **Addressing Reward Sparsity & Exploration:**
 - Clip-Higher (exploration), Token-level Loss (normalization), Positive Example LM Loss (self-imitation), Group Sampling (baseline estimation)

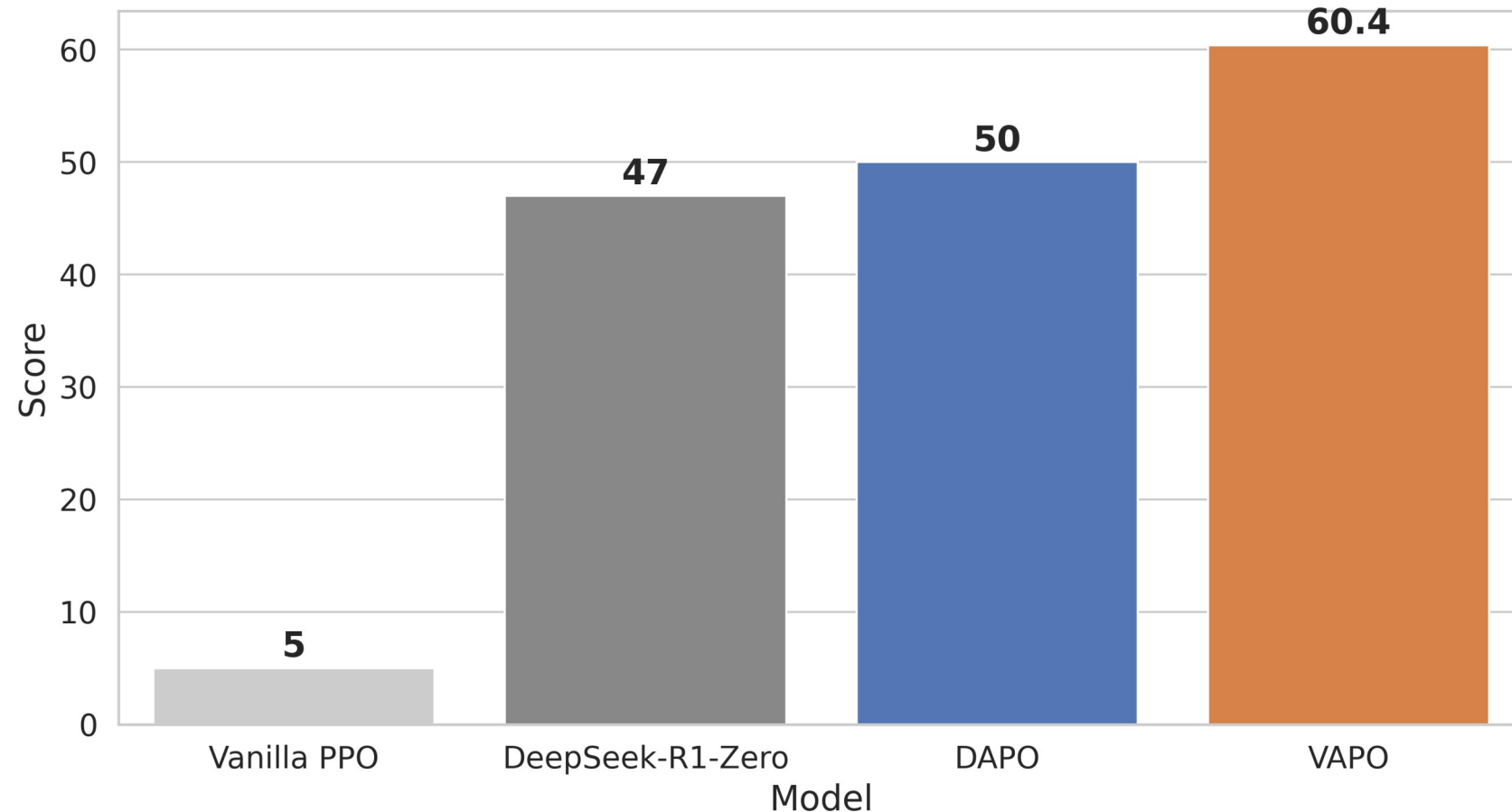
Length-Adaptive GAE: Dynamic λ Adjustment

- A novel technique to optimize the bias-variance tradeoff across heterogeneous sequence lengths:
- **Dynamic Adjustment Mechanism:**
 - The λ parameter in Generalized Advantage Estimation (GAE) dynamically shifts based on response length.
- **Short Responses = Lower λ (Lower Variance):**
 - Prioritizes variance reduction where bias accumulation is less of a concern.
- **Long Responses = Higher λ (Lower Bias):**
 - Reduces bias in long reasoning chains where bootstrapped value estimates can compound errors.

VAPO Scores 60.4 on AIME 2024

— 10+ Points Above DAPO

AIME 2024 avg@32 Performance

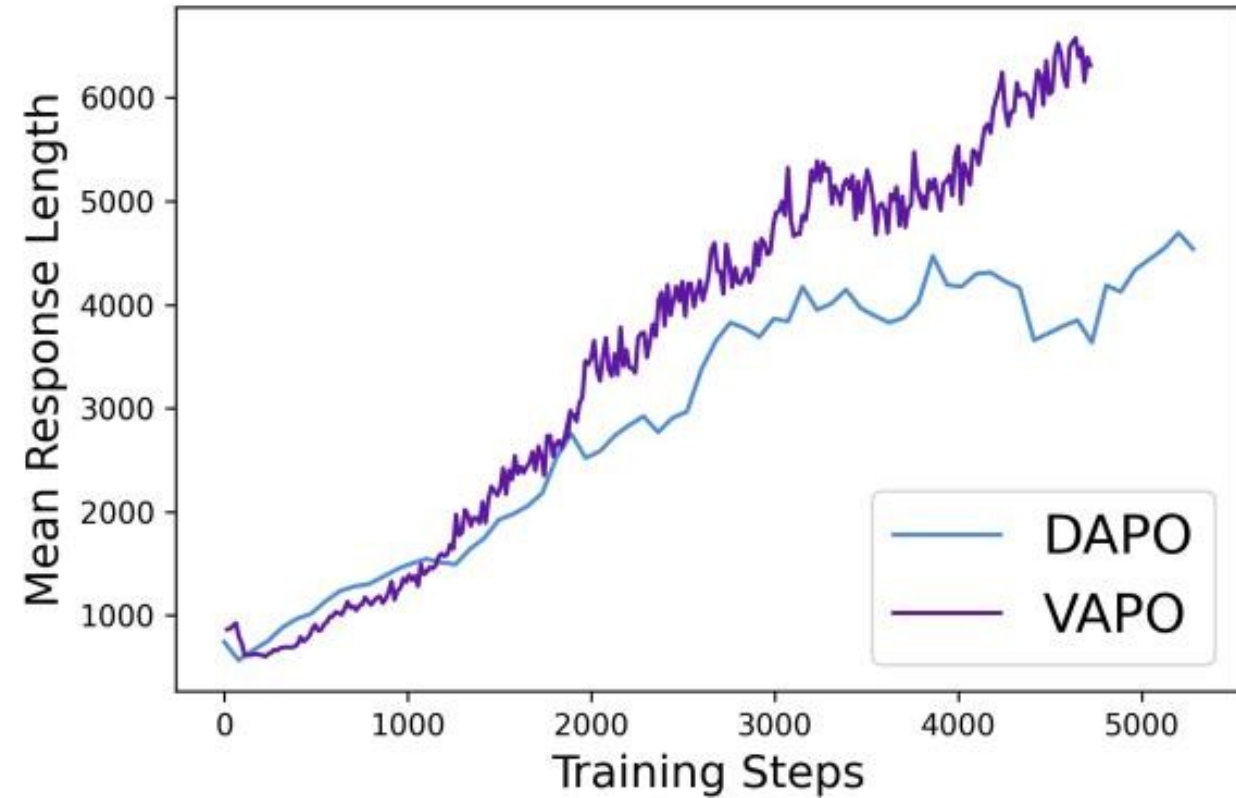
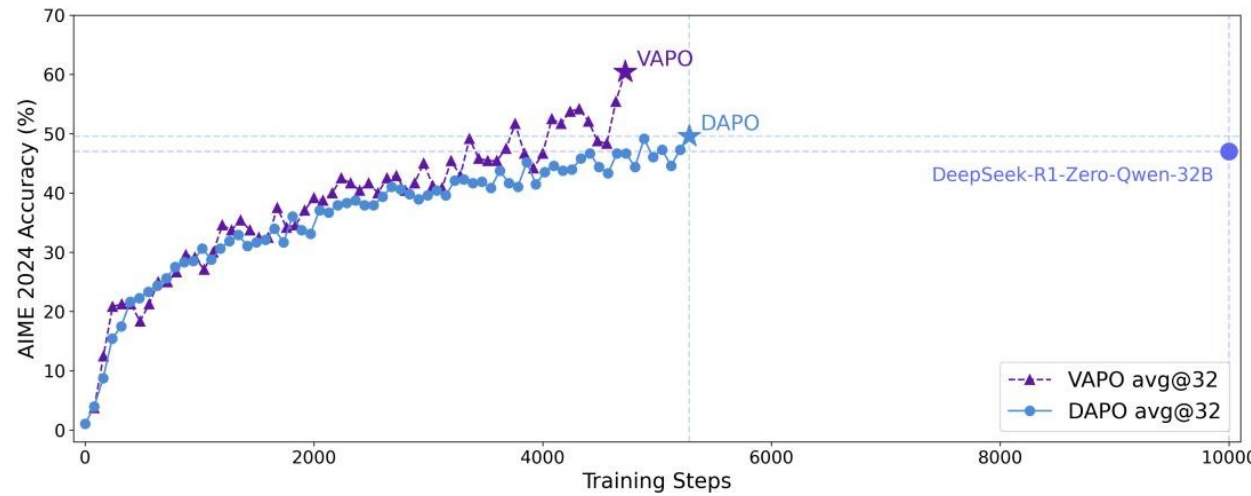


Key Results:

- Achieves 60.4 avg@32 on AIME 2024.
- Surpasses DAPO (50) by 10.4 points.
- Outperforms DeepSeek-R1-Zero-Qwen-32B (47).
- Evaluated on identical Qwen2.5-32B base model with no SFT data.
- Refer to Figure 2 for the training curve demonstrating consistent superiority throughout training.

VAPO Reaches SOTA in 5K Steps

— Half the Training Budget



Highly Efficient & Stable Training:

- VAPO reaches ~60% accuracy around step 4,500, while DAPO plateaus near 50% at step 5,500.
- DeepSeek-R1-Zero-Qwen-32B requires ~10,000 steps to reach just 47%.
- VAPO generates longer, more elaborate reasoning chains (reaching ~6,000 tokens vs DAPO's ~4,500).
- Zero training crashes across multiple independent runs.

Ablation: Every Component Matters — Value-Pretraining

Value-Pretraining

Component Removed	Score	Drop from 60
None (Full VAPO)	60	-
w/o Group-Sampling	55	-5
w/o Positive Example LM Loss	54	-6
w/o Token-level Loss	53	-7
w/o Clip-Higher	46	-14
w/o Length-adaptive GAE	45	-15
w/o Decoupled-GAE	33	-27
w/o Value-Pretraining	11	-49

Critical Insights:

- Value-Pretraining is the single most vital component; removing it drops the score from 60 to 11.
- Decoupled-GAE and Length-Adaptive GAE also drive significant performance gains.

Key Takeaways: Value-Model-Based RL Is Back

- **Value-Model-Based Methods Can Outperform Value-Free RL**
 - They succeed in complex reasoning when challenges like bias, heterogeneous lengths, and reward sparsity are effectively addressed.
- **Length-Adaptive GAE Is Critical for Long Chains**
 - A key innovation driving a 15-point increase by optimizing the bias-variance tradeoff.
- **Value-Pretraining Is Non-Negotiable**
 - Without it, value-model-based RL barely functions (contributes 49 points).